

Student Number

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2019

**TRIAL HIGHER SCHOOL
CERTIFICATE
EXAMINATION**

PEM

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your student number in the space provided

Total marks – 100

100 marks

This exam has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1-20
- Allow about 35 minutes for this part

Part B – 80 marks

- Attempt Questions 21-34
- Allow about 2 hour and 25 minutes for this part

Directions to School or College

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Part A – 20 marks

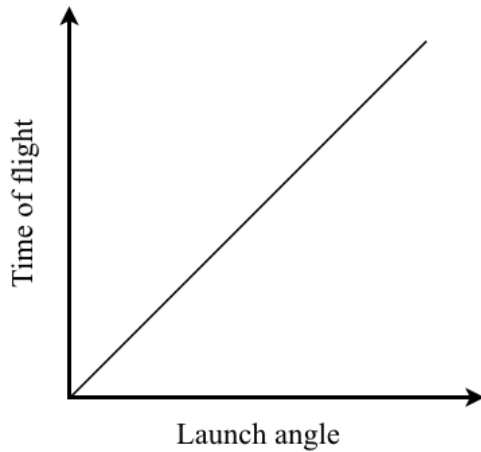
Attempt Questions 1-20

Allow about 35 minutes for this part

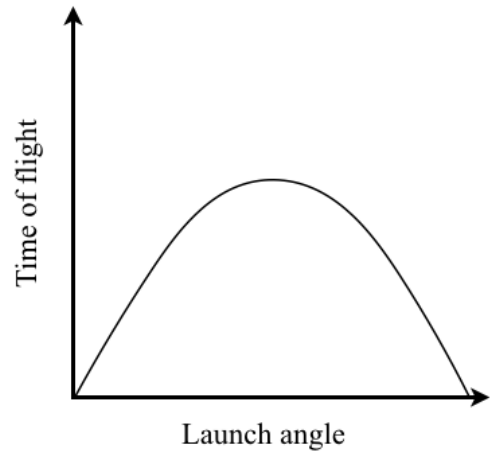
Use the multiple choice answer sheet for Questions 1-20

- 1 Which graph below correctly represents the relationship between launch angle and time of flight for a projectile?

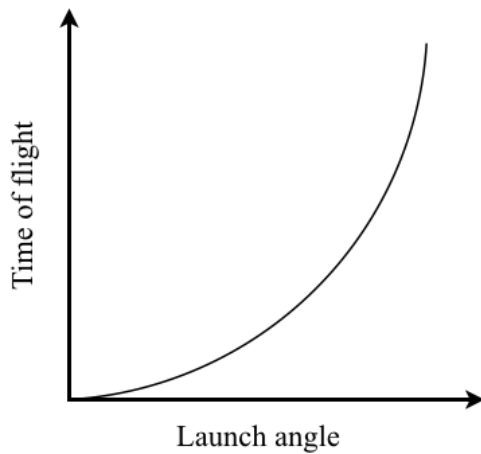
(A)



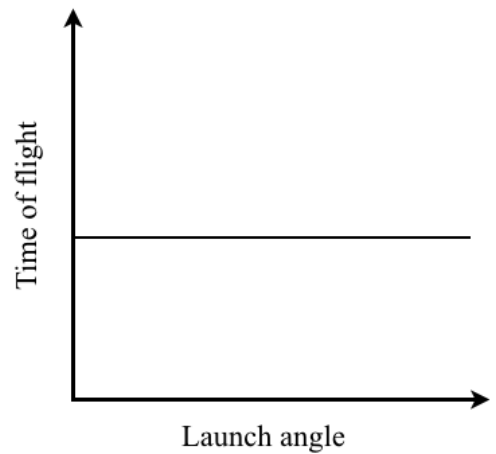
(B)



(C)



(D)

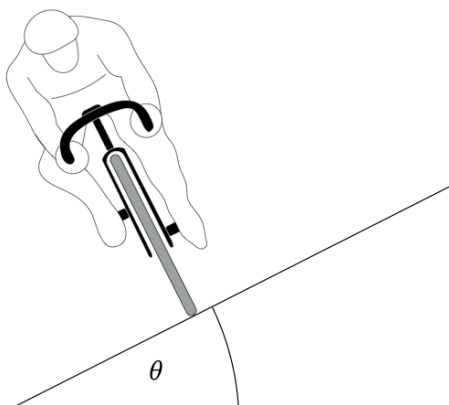


- 2 Using a homemade projectile launcher a student undertook a depth study to analyse projectile motion.

The student determined that the launcher did not produce a consistent initial velocity.

The type of error this produced would be best defined as:

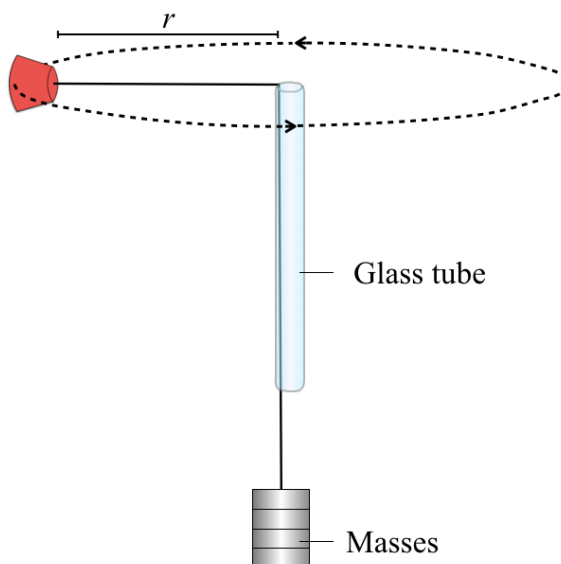
- (A) Instrumental error
 - (B) Human error
 - (C) Systematic error
 - (D) Random error
- 3 The diagram below shows a cyclist turning on a banked track.



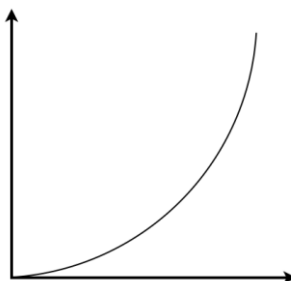
Which of the following statements about the forces acting on the cyclist are correct?

- (A) All the forces are balanced, allowing him to travel at a constant speed.
- (B) The net force acting on the cyclist creates a centripetal force.
- (C) The force acting on the cyclist parallel to the bank is balanced by the centripetal force.
- (D) The component of the normal force perpendicular to the slope is equal to the weight force.

- 4 A class undertook a depth study in pairs to investigate centripetal force using the apparatus below.



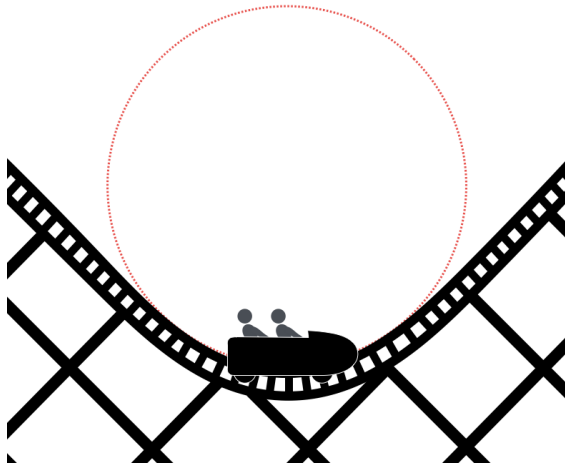
One day a student was sick and their partner had to interpret their results. The lab partner found a graph with unlabelled variables as shown below.



Which pair of variables was most likely graphed?

- (A) F and v
 - (B) F and r
 - (C) v and r
 - (D) a and v/r
- 5 Thompson's charge-to-mass experiment:
- (A) calculated the mass of the electron.
 - (B) measured the charge of an electron.
 - (C) provided evidence that cathode rays had mass.
 - (D) demonstrated that cathode rays travelled in straight lines.

- 6 A rollercoaster trolley with a constant velocity on a dip has a centripetal acceleration due to the circular path, as shown in the diagram.

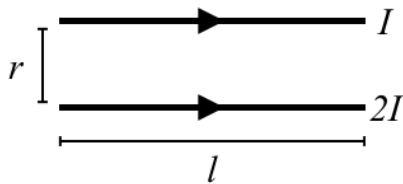


Which of the following descriptions best analyses the forces acting on the trolley?

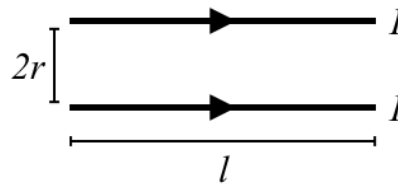
- (A) The gravitational force acting on the trolley is larger than when travelling on a flat surface.
- (B) The centripetal force is equal to the normal force.
- (C) The sum of the Normal force and the gravitational force is equal to the centripetal force.
- (D) The normal force is larger than when travelling on a flat surface because the centrifugal force is larger.

- 7 Which of the following pairs of current carrying conductors has the weakest force acting between them?

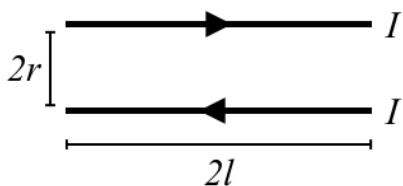
(A)



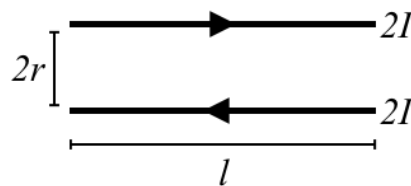
(B)



(C)

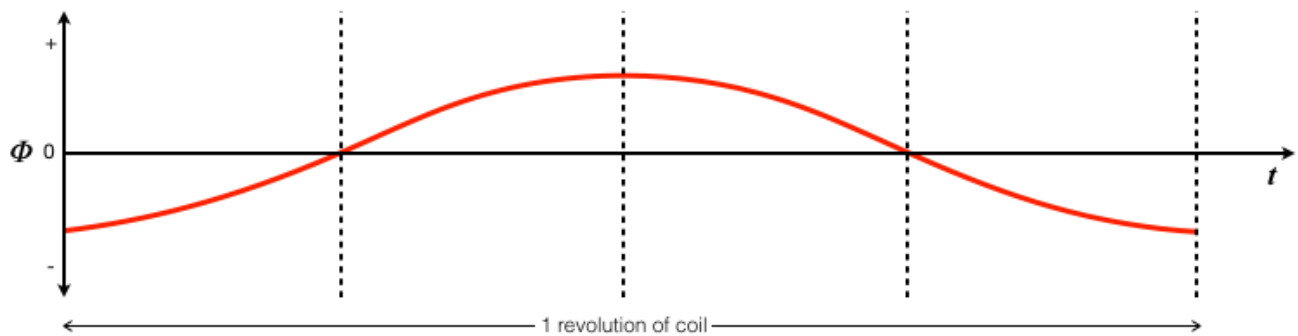


(D)



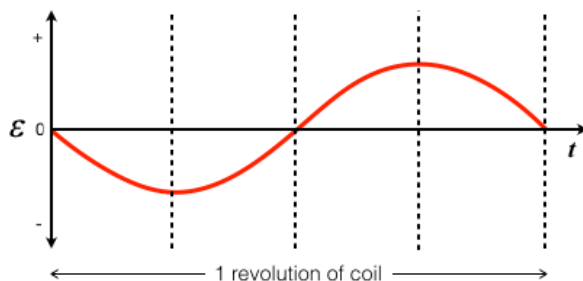
Use the following information to answer Question 8 and Question 9

The graph below shows the variation of flux of a generator coil as it completes a single revolution.

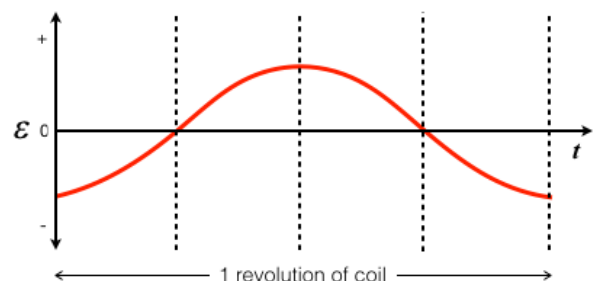


8 Which of the following graphs shows the corresponding EMF in the generator?

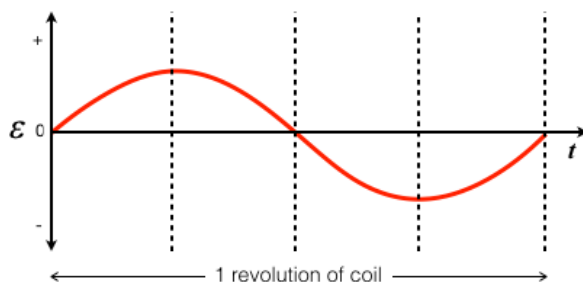
(A)



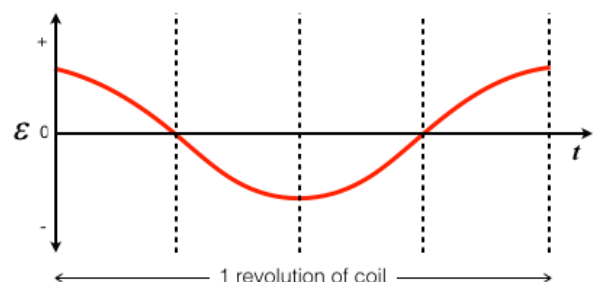
(B)



(C)

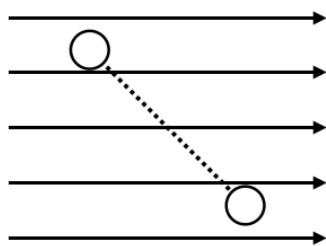


(D)

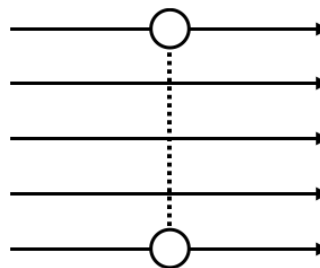


- 9 The following diagrams represent a cross section of the generator coil. Which position was the coil in at the beginning of the graph?

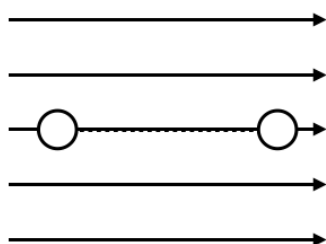
(A)



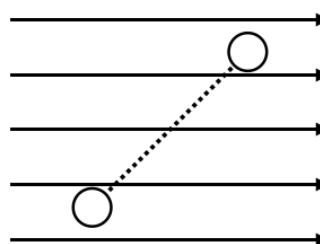
(B)



(C)



(D)



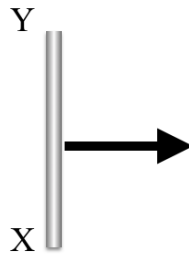
- 10 The main difference between a controlled and an uncontrolled nuclear chain reaction is:

- (A) the number of neutrons released at each fission.
- (B) the amount of energy released at each fission.
- (C) the amount of subsequent fission reactions.
- (D) the rate of fission reactions.

- 11 Which of the following formulas shows the correct decay of Cobalt-60 to Nickel-60?

- (A) ${}_{27}^{60}\text{Co} \rightarrow {}_{28}^{60}\text{Ni} + {}_{-1}^0\beta + \bar{\nu}$
- (B) ${}_{27}^{60}\text{Co} \rightarrow {}_{25}^{56}\text{Ni} + {}_2^4\text{He} + \text{energy}$
- (C) ${}_{27}^{60}\text{Co} \rightarrow {}_{28}^{60}\text{Ni} + {}_{+1}^0\beta + \nu$
- (D) ${}_{27}^{60}\text{Co} \rightarrow {}_{27}^{60}\text{Ni} + {}_{+1}^0\beta + \bar{\nu}$

- 12 A wire conductor is moving to the right as shown below.

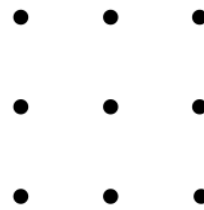


Which of the following magnetic fields would the conductor need to pass through in order to induce a current in the direction X to Y in the conductor?

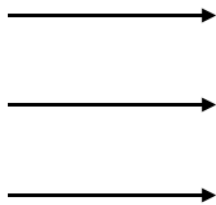
(A)



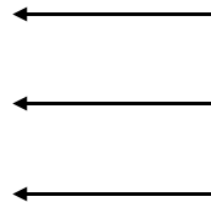
(B)



(C)



(D)



- 13 Foucault measured the speed of light by focusing a light onto a rotating mirror which reflected it onto a fixed mirror which in turn reflected it back. Whilst the light travelled to the fixed mirror and back, the rotating mirror rotated through an angle. By measuring the angle with a known angular velocity of the mirror, Foucault was able to calculate a value for the speed of light.

The smaller the measured angle the more uncertainty there was in the measurement.

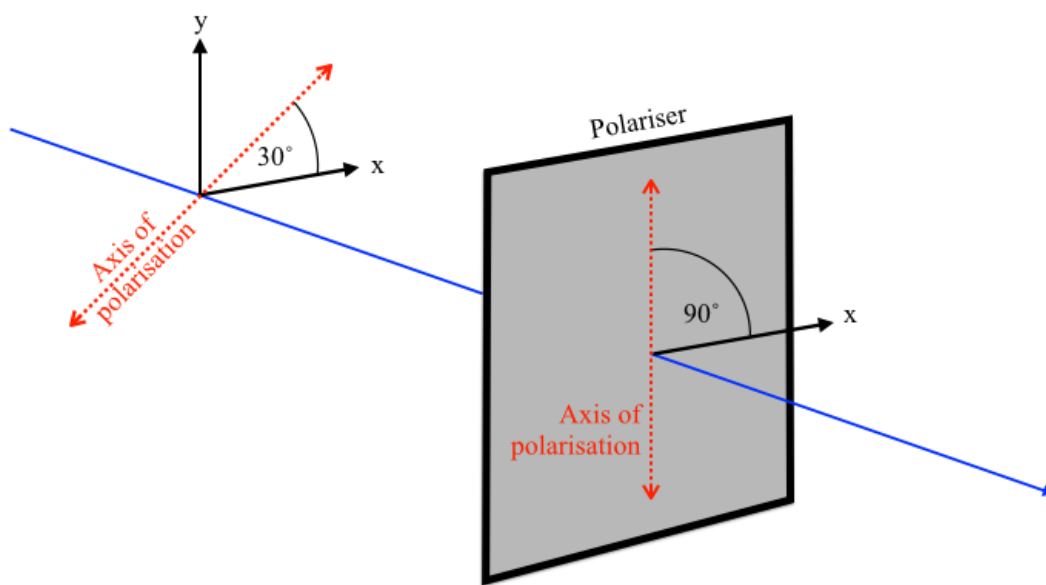
What would be the easiest adjustment to make to the experimental design to increase the size of the angle and reduce the uncertainty in measurement?

- (A) Increase the distance between the mirrors.
- (B) Decrease the distance between the mirrors.
- (C) Increase the rotational velocity of the rotating mirror.
- (D) Decrease the rotational velocity of the rotating mirror.

14 Calculate the energy released from an electron-positron annihilation.

- (A) $1.638 \times 10^{-13} \text{ eV}$
- (B) $8.19 \times 10^{-14} \text{ eV}$
- (C) 0.51 MeV
- (D) 1.02 MeV

15 A polarised light source is entering a polarising filter as shown below.



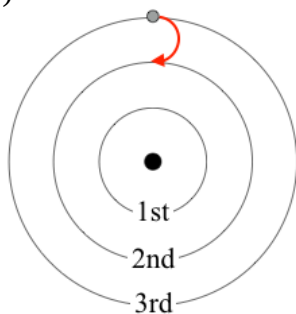
What adjustment would produce the greatest reduction in light intensity?

- (A) Rotating the light source clockwise 10°
 - (B) Rotating the light source counter clockwise 70°
 - (C) Rotating the filter clockwise 130°
 - (D) Rotating the filter counter-clockwise 110°
- 16 Calculate the work function for copper, which has a threshold frequency of $1 \times 10^9 \text{ MHz}$.
- (A) 4.1 eV
 - (B) 4.1 MeV
 - (C) 3.9 eV

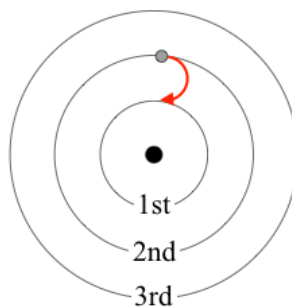
(D) 3.9 MeV

- 17 Which of the following examples below of an electron moving energy levels would produce the shortest wavelength?

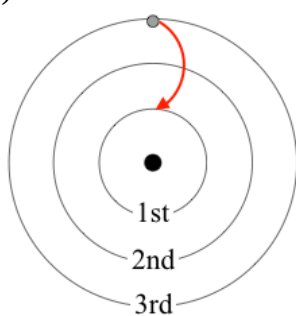
(A)



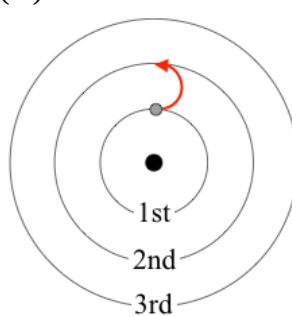
(B)



(C)



(D)



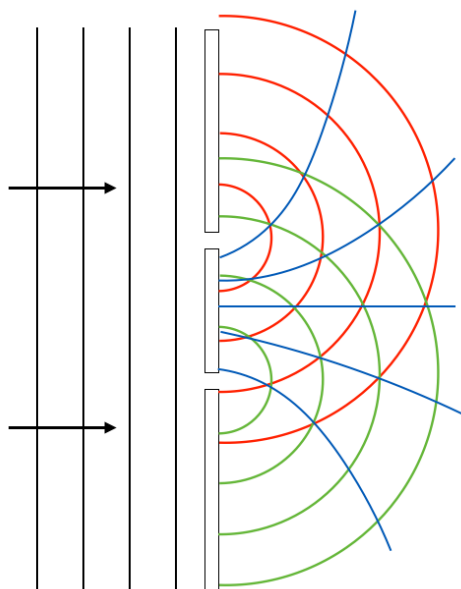
- 18 What significant change to the atomic model did the Gold foil experiment lead to?

- (A) All atoms contained negative electrons.
- (B) Electrons existed in energy shells surrounding the nucleus.
- (C) The nucleus was a positive space.
- (D) The atom is mostly empty space.

- 19 Calculate the wavelength of a tennis ball of mass 59.4g travelling at 170 km h⁻¹.

- (A) 6.56×10^{-38} m
- (B) 2.36×10^{-37} m
- (C) 6.56×10^{-35} m
- (D) 2.36×10^{-34} m

20 What phenomenon is best demonstrated in the experiment shown in the diagram below?



- (A) Wave diffraction
- (B) Wave interference
- (C) Wave resonance
- (D) Wave refraction

Part B – 80 marks

Attempt questions 21-40

Allow about 2 hour and 25 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (2 marks)

Low Earth orbit satellites experience significant orbital decay, have a low life span and are difficult to monitor and troubleshoot. Explain why low Earth orbit satellites are still used extensively despite the above points.

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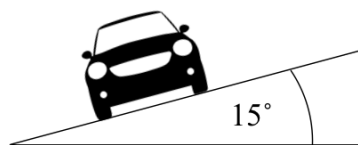
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Question 22 (5 marks)

- (a) A car is travelling at the design speed of a banked track with a turning radius of 26.0 m, as shown in the diagram. The centripetal force acting on the car is 3040 N.

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Determine the mass of the car.

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- (b) Explain, in terms of forces, why a car would slide off a frictionless banked track if it exceeded the design speed. 3

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Question 23 (3 marks)

- In terms of mechanical energy, describe why the work done on an object in uniform circular motion is equal to zero on a horizontal plane. 3

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Question 24 (3 marks)

Explain quantitatively and qualitatively why the mass of an object is not a factor in determining its escape velocity.

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Question 25 (5 marks)

Maverick's F18 plane is flying at a constant altitude of 200.0 m and a speed of 300.0 ms^{-1} . His rockets accelerate at 25.0 ms^{-2} for 10.0 seconds in a straight line until the fuel runs out. At this point the missiles follow normal projectile motion and cruise down to the surface.

- (a) Calculate the distance travelled by the rockets whilst they are using fuel. 2

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- (b) Determine the distance from his target that Maverick should fire his missile in order to hit it. 3

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Question 26 (2 marks)

The Earth's average distance from the Sun is 1.00 AU. The average distance of Saturn from the Sun is 9.00 AU. Calculate the period of Saturn.

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Question 27 (3 marks)

Justify Lenz's Law.

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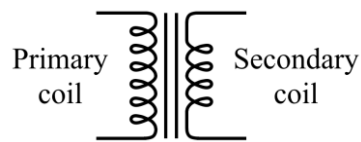
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Question 28 (3 marks)

Explain how the device shown in the diagram below works.

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Question 29 (4 marks)

A power plant generates 150 MW of electricity for a town 20 km away. The transmission wires have a resistance of 0.01 W per km. The voltage drop between the plant and the town is 50 V.

- (a) Calculate the power loss between the plant and the town

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- (b) Calculate the voltage transmitted by the plant.

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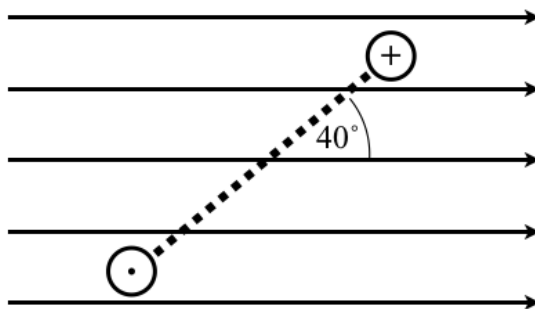
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Question 30 (5 marks)

Below is a diagram of a cross section of a coil in a magnetic field. The square coil of side length 12.00 cm has 8 turns, and a current of 3.00 A. The magnetic field has a strength of 2.50×10^4 T. What is the magnitude of the torque on the coil when it is in the position shown below.



- (a) Calculate the torque on the coil.

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- (b) Determine the change in flux if it continues to rotate from the above position 30° .

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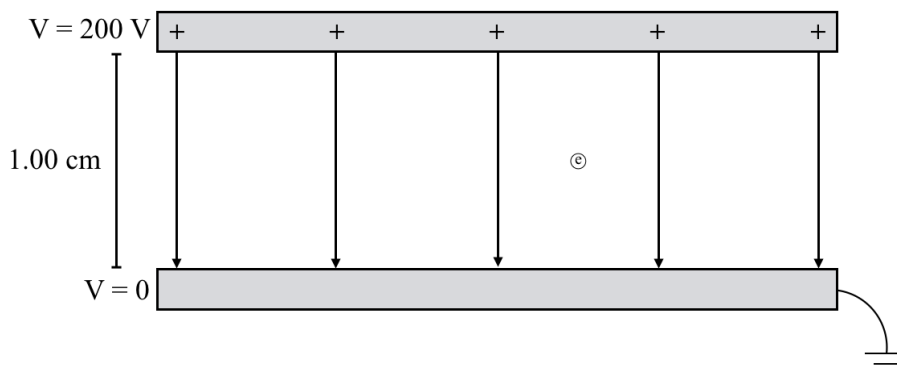
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Question 31 (5 marks)

Two parallel plates are set up as shown in the diagram below so that the positive plate is above the earthed plate.



An electron is placed exactly between the plates so that it is allowed to fall.

- (a) Calculate the net force acting on the electron.

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- (b) Determine how long it takes for the electron to touch a plate.

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Question 32 (6 marks)

Discuss the significance of Maxwell's contribution to the development of our current understanding of electromagnetism.

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[illegible]

Question 33 (6 marks)

- (a) Describe the evidence that supported Newton and Huygens proposed models of light. **4**

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- (b) Describe why the wave theory of light was inadequate at explaining the photoelectric effect. **2**

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Question 34 (5 marks)

- (a) A spaceship pilot travelling through a spaceport observes that it takes 2.00 s to pass through. A person viewing the spaceship from the spaceport records the ship taking 4.39 s. Determine the speed of the ship as a percentage of the speed of light. **3**

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- (b) Outline an experimental validation of time dilation. **2**

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Question 35 (6 marks)

- (a) Account for the production of absorption spectra from stars.

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- (b) Explain how rotational velocity can be deduced from spectra.

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Question 36 (3 marks)

Describe Hubble's justification for an expanding universe.

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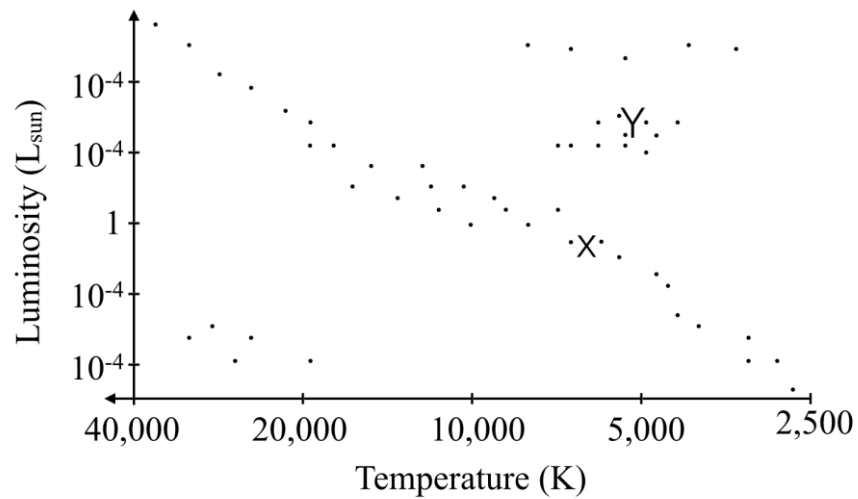
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Question 37 (4 marks)

Identify and outline the most likely dominant nucleosynthesis reactions occurring in the stars marked X and Y in the H-R diagram below.

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[illegible]

Question 38 (5 marks)

- (a) Outline the insight that Bohr achieved from analysing the emission spectra of hydrogen. **2**

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- (b) Calculate the second longest wavelength of the Balmer series in hydrogen. **3**

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Question 39 (4 marks)

A sample of Iodine-131 has a half-life of 8 days and contains 1.2×10^{13} nuclei.

- (a) Calculate the decay constant for Iodine-131.

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- (b) Determine how many Iodine-131 nuclei will remain after 76 days.

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Question 40 (6 marks)

Describe the electromagnetic composition of Baryons.

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